

# A Hybrid Matrix Completion + Liquid Neural Network Framework for Groundwater Level Imputation

A General Framework Coupling Spatial Correlation with Continuous-Time Dynamics for Sparse Well Networks

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## 1 The Problem

Groundwater monitoring networks worldwide contain **millions of measurements** but are plagued by **long temporal gaps** spanning months to years. Jasechko et al. (2024) found rapid declines in 30% of 1,693 aquifer systems globally, yet most wells lack continuous records needed for trend analysis.

>5yr

Typical Max Gap

<50

Median Obs/Well

0.3

Prior KGE at 2yrs

Prior methods (ELM, LSTM, GRU) treat each well independently in the temporal domain. Matrix completion has been applied only once to groundwater (Sharma et al., 2024) as a standalone method. No prior work couples spatial MC with temporal neural dynamics.

## 2 MC+LNN Pipeline

### Stage 1: PCHIP

Fill gaps ≤ 24 months  
Shape-preserving  
polynomial interpolation

### Stage 2a: ARCHI

Top-15 correlated donor  
wells via OLS regression  
Trend-aware initialization

### Stage 2b: MC

Softimpute SVD on  
composite matrix:  
donors + GLDAS + sin/cos

### Stage 2c: LNN

CFC reservoir dynamics  
MC as input placeholder  
Trained on observations only

**Key insight:** MC provides **spatial context** (cross-well correlations), LNN provides **temporal dynamics** (nonlinear continuous-time evolution). MC predictions drive the LNN reservoir, but readout trains only on real observations.

$$\text{LNN CFC: } x(t+dt) = x \cdot e^{-\text{leak} \cdot dt} + (b / \text{leak}) \cdot (1 - e^{-\text{leak} \cdot dt}) \quad \text{where } b = \tanh(W_{in} \cdot \text{input} + W_{res} \cdot x)$$

## 4 Cross-Validation Results (Validated on 592 wells, Great Salt Lake Basin)

### A. Random Missing Data (50 trials each)

| % Removed | KGE           | R²    | RMSE (ft) | MAE (ft) |
|-----------|---------------|-------|-----------|----------|
| 5%        | 0.837 ± 0.085 | 0.771 | 2.53      | 1.98     |
| 10%       | 0.843 ± 0.077 | 0.770 | 2.70      | 2.06     |
| 20%       | 0.853 ± 0.051 | 0.787 | 2.64      | 1.98     |
| 30%       | 0.844 ± 0.048 | 0.770 | 2.79      | 2.03     |
| 40%       | 0.851 ± 0.030 | 0.784 | 2.75      | 2.03     |
| 50%       | 0.847 ± 0.035 | 0.788 | 2.74      | 2.03     |

KGE remains >0.84 even when 50% of data is removed.

### B. Consecutive Year Gaps (20 trials each)

| Gap Size | KGE           | R²    | RMSE (ft) | MAE (ft) |
|----------|---------------|-------|-----------|----------|
| 1 year   | 0.783 ± 0.116 | 0.703 | 2.98      | 2.28     |
| 2 years  | 0.792 ± 0.081 | 0.721 | 2.97      | 2.29     |
| 3 years  | 0.802 ± 0.076 | 0.730 | 3.02      | 2.33     |
| 4 years  | 0.806 ± 0.055 | 0.738 | 2.96      | 2.15     |
| 5 years  | 0.815 ± 0.063 | 0.744 | 2.91      | 2.20     |

KGE >0.78 even for 5-year consecutive gaps.

## 3 What's New

First framework to jointly exploit:

### Spatial Structure

Matrix Completion exploits correlation across the full 592-well network via low-rank SVD with GLDAS auxiliary rows and seasonal encoding

### Temporal Dynamics

Liquid Neural Network with closed-form continuous-time cells (Hasani et al., 2022) captures nonlinear drought response, recharge lags, and seasonal patterns

vs. Prior approaches:

- **ELM** (Evans et al., 2020): Temporal only, no cross-well structure
- **LSTM/GRU** (Jeong et al., 2020; Wunsch et al., 2021): Each well independent
- **Softimpute** (Sharma et al., 2024): Spatial only, no temporal model
- **MC+LNN (ours)**: Spatial + Temporal, coupled

## 5 PCHIP Matters

Replacing LNN-based small-gap fill with PCHIP in Stage 1 improves performance across the board:

Random Missing (avg KGE)



Consecutive Gaps (avg KGE)



+10%

Random KGE

+19%

Consec. KGE

PCHIP densification improves donor correlations for the MC stage, cascading through the pipeline.

## 6 Each Stage Adds Value

Ablation study isolating each component:

| Method    | KGE (random) | KGE (consec.) |
|-----------|--------------|---------------|
| MC alone  | 0.24         | -0.07         |
| LNN alone | 0.65         | 0.45          |
| MC + LNN  | 0.85         | 0.80          |

MC provides spatial placeholders. LNN refines them with temporal dynamics. **Neither alone achieves what the coupled framework delivers.**

## 7 General Framework, Validated on GSLB

Applicable to Any Well Network

- **Inputs:** Well locations, irregular observations, GLDAS auxiliary data (globally available)
- **No basin-specific tuning:** Donor selection, MC rank, and LNN hyperparameters are auto-optimized per well
- **Scales:** Tested on 592 wells; pipeline handles any network size
- **Handles heterogeneity:** WTE range 4,200-7,200 ft across GSLB; per-row normalization adapts to any range
- **Open source:** Runs in browser (TypeScript) or Python backend

### Validation Basin: GSLB

Area: 93,000 km²  
Wells: 592 (2000-2023)  
Months: 288  
Aux: 5 GLDAS soil moisture vars  
Donors: top 15 by correlation  
MC ranks: 3, 5, 8, 10, 12 (auto)  
LNN ensemble: 3 models  
Result: 100% completeness

## 8 Conclusions

- **KGE > 0.84** under random missing data from 5-50%, with remarkable consistency across missing-data rates
- **KGE > 0.78 for 1-5 year consecutive gaps** -- the hardest imputation scenario where prior methods degrade to KGE < 0.3
- **PCHIP + MC + LNN coupling is essential:** MC alone (0.24) and LNN alone (0.65) both underperform the integrated pipeline (0.85)
- **General-purpose:** No basin-specific tuning required. Uses globally available GLDAS auxiliary data and auto-optimized parameters
- **Scalable:** Validated on 592 wells; runs in browser or Python

Applicable to any groundwater monitoring network with sparse, irregular observations and multi-year gaps



